expansion in volume is termed as **volume expansion**. It is shown in Figure 8.4

Linear Expansion

In solids, for a small change in temperature ΔT, the fractional change in length $\left(\frac{\Delta L}{L_0}\right)$ is directly proportional to ΔT.

$$\frac{\Delta L}{L_0} = \alpha_L \Delta T$$

Therefore, $\alpha_L = \frac{\Delta L}{L_0 \Delta T}$

Where, α_L = coefficient of linear expansion.

ΔL = Change in length

L_0 = Original length

ΔT = Change in temperature.



- When the lid of a glass bottle is tight, keep the lid near the hot water which makes it easier to open. It is because the lid has higher thermal expansion than glass.
- When the hot boiled egg is dropped in cold water, the egg shell can be removed easily. It is because of the different thermal expansions of the shell and egg.

EXAMPLE 8.6

Eiffel tower is made up of iron and its height is roughly 300 m. During winter season (January) in France the temperature is 2°C and in hot summer its average temperature 25°C. Calculate the change in height of Eiffel tower between summer and winter. The linear thermal expansion coefficient for iron α = 10 ×10^{-6} per °C



Solution

$$\frac{\Delta L}{L_0} = \alpha_L \Delta T$$

$$\Delta L = \alpha_L L_0 \Delta T$$

$\Delta L = 10 \times 10^{-6} \times 300 \times 23 = 0.069$ m=69 mm

Area Expansion

For a small change in temperature ΔT the fractional change in area $\left(\frac{\Delta A}{A_0}\right)$ of a substance is directly proportional to ΔT and it can be written as

$$\frac{\Delta A}{A_0} = \alpha_A \Delta T$$

Therefore, $\alpha_A = \frac{\Delta A}{A_0 \Delta T}$

Where, α_A = coefficient of area expansion.

ΔA = Change in area

A_0 = Original area

ΔT = Change in temperature

Volume Expansion

For a small change in temperature ΔT the fractional change in volume $\left(\frac{\Delta V}{V_0}\right)$ of a substance is directly proportional to ΔT.

$$\frac{\Delta V}{V_0} = \alpha_V \Delta T$$

Therefore, $\alpha_V = \frac{\Delta V}{V_0 \Delta T}$

UNIT-8(XI-Physics_Vol-2).indd 103 02-01-2023 15:25:27